

Eric (Chen-Chien) Chang

BACKEND / DATA ENGINEER

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PROFILE

Backend and data engineer building reliable pipelines and distributed systems for complex data workflows. Full-time independent open-source contributor; Apache Kafka contributor and Apache Gravitino committer with hands-on work in stream processing, database performance, observability, and real-world data federation.

EXPERIENCE

Open Source For You

2024 – Present

Independent Open Source Contributor (full-time since 2025)

Technical Community

- **Apache Kafka Contributor** — authored [KIP-1320](#) with a draft PoC ([PR #22758](#)); contributing a commit-before-partition-switch fix ([PR #21584](#)) and bounded Streams buffering ([PR #22458](#)).
- **Community Over Code Asia 2026 Speaker** — [Kafka Streams DLQ and exactly-once safety \(session\)](#).
- **Apache Gravitino Committer** — made PostgreSQL/MySQL integration tests reproducible, cut the Hive test image by **420MB**, and completed Python partition APIs.

Lawsnote

Jan 2024 – Dec 2024

Backend Software Engineer

Taipei, Taiwan

- Streamed CPU-intensive analysis results, letting users see output in **5s instead of waiting 100s**.
- Built a monitoring pipeline that cut detection time for data loss and unhealthy remote agents from **22 days to 1 day**.
- Removed a cross-language bottleneck in a Node.js native module through profiling.
- Converted ambiguous regulatory text into structured data and automated government-act refreshes.

MediaTek (Contract)

Feb 2022 – Mar 2023

Backend Software Engineer

Hsinchu, Taiwan

- Cut engineers' data-warehouse UI wait from **60s to 1s** by redesigning its PostgreSQL access path ([case study](#)).
- Cut simulation-data processing from **2hr to 1min**, shortening engineering feedback cycles.
- Kept long-running simulations reliable with a queue-backed service across a **5-node** cluster.
- Cut Go service memory use by **50%** and added telemetry to diagnose failures and resource pressure.

National Cheng Kung University

Sep 2017 – Jul 2019

Research Assistant

Tainan, Taiwan

- Conducted FDM 3D-printing research and built experimental tooling with Python, C, MicroPython, and Arduino.

SELECTED PRODUCT

Locus — Real-World Context Federation for AI Agents

2026 – Present

Independent Technical Project

MCP / Kafka / RisingWave / Iceberg / DuckDB / Cloudflare

- Built and launched Locus, giving AI clients one MCP interface over **2,300+ spatial entities from 13 source IDs**, including **1,800+ parking facilities**, **500+ EV-charging stations**, and live coach telemetry.
- Designed a provider-neutral semantic layer for entities, observations, and relationships that preserves freshness, provenance, coverage gaps, and conflicts.
- Built a streaming lakehouse for minute-level coach telemetry (**about 28,800 observations/day**) with Kafka, RisingWave, Iceberg, and DuckDB, separating online serving from analytics, replay, and audit.

TECHNICAL SKILLS

Languages: Java, Go, JavaScript / TypeScript, Python, SQL

Data & Streaming: Apache Kafka, Kafka Streams, PostgreSQL, RisingWave, Apache Iceberg, DuckDB

Platforms & Observability: Node.js, Docker, Cloudflare Workers, OpenTelemetry, Model Context Protocol (MCP)

EDUCATION

National Taiwan Ocean University — Electrical Engineering coursework, degree not completed

2013 – 2017